Jeroen Van Goey

Staff Research Engineer in BioAI building generative models for biology

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Summary

Machine learning research engineer working at the intersection of AI and science, bridging scientific ML, biological domain knowledge, production-quality software, and team leadership. I build generative models and take them from prototype to deployment, with a focus on *de novo* peptide sequencing.

8 publications, including papers published in *Nature Machine Intelligence* and *Nature Communications*.

Currently looking for Staff/Principal Research Engineer roles in AI for science. Based in Cape Town, open to remote-first roles now and hybrid/on-site roles in Europe or the USA from 2027.

Experience

InstaDeep, Staff Research Engineer · BioAI Lead — Cape Town, South Africa — Aug 2022 – present — 4 years 1 month

- **Tech lead and hiring manager** for two cross-disciplinary **ML research teams** of about 5 to 8 people: *de novo* peptide sequencing (with the Technical University of Denmark) and signal-peptide design for secretion efficiency (with BioNTech).
- Delivered a **client collaboration with Syngenta**, applying **genomic language models** (AgroNT, a Nucleotide Transformer trained on ~10.5 million genomic sequences spanning trillions of base pairs across 48 plant species) to accelerate crop trait research.
- Set **research direction** and take models **from research to production** (InstaNovo is available to commercial customers on DeepChain), balancing scientific rigour with reliable engineering.
- **Lead the BioAI team** of InstaDeep's Cape Town office and serve as the office's **site manager** (an office of about 25 people); hire and grow the BioAI teams across the Cape Town and Kigali offices.
- Directed the team's **ML engineering foundations**: scalable, Transformer-based libraries (Python / PyTorch) for large-scale training on InstaDeep's Kyber (~500 PFLOPs) and EuroHPC's MareNostrum 5 (260 PFLOPs) supercomputers, and the cloud (AWS, GCP).

Barco, Senior Software Development Engineer, Machine Learning — Kortrijk, Belgium — Feb 2020 – Aug 2022 — 2 years 7 months

- Built a TensorFlow Extended production pipeline (orchestrated with Apache Airflow) training deep-learning models on multispectral images to detect and classify melanoma skin cancers for Demetra, Barco's dermatology imaging device.
- Engineered the pipeline with **end-to-end data and model lineage tracking** to deliver the traceability and reproducibility that **regulated medical industries (FDA approval)** require.
- Worked across the research, cloud-backend and mobile/web frontend teams.

Bayer Crop Science, then BASF, Bioinformatics Researcher, Manager of the Python & R Platforms — Zwijnaarde, Belgium — Feb 2018 – Jan 2020 — 2 years

- Owned the **Python/R data-analysis platform** used by **480 researchers and data scientists**: technical support, proactive monitoring and root-cause analysis, **training** (NumPy, pandas, BioPython) and **mentoring** across the company's internal research community.
- The Zwijnaarde site was divested from Bayer Crop Science to BASF as part of Bayer's acquisition of Monsanto; I continued in the same role across the transition.

Applied Maths (acquired by bioMérieux), Bioinformatics Software Developer — Sint-Martens-Latem, Belgium — Sept 2011 – Jan 2018 — 6 years 5 months

- Worked on BioNumerics, a bioinformatics suite for integrated analysis of biological data, writing **custom Python scripts and extensions** for clients in the clinical, academic and government sectors.

Selected publications

InstaNovo enables diffusion-powered *de novo* peptide sequencing in large-scale proteomics experiments — Nature Machine Intelligence — 2025
K. Eloff, K. Kalogeropoulos, …, **J. Van Goey**, …, T. P. Jenkins
Trained on ~63 million spectra; top-ranked in an independent benchmark of 17 de novo tools across 83 datasets; Altmetric attention score 100+ and 120+ GitHub stars.
10.1038/s42256-025-01019-5

InstaNovo-P: a *de novo* peptide sequencing model for phosphoproteomics — Nature Communications — 2026
J. Lauridsen, P. Ramasamy, …, **J. Van Goey**, …, K. Kalogeropoulos
10.1038/s41467-026-75138-x

***De novo* peptide sequencing rescoring and FDR estimation with Winnow** — arXiv 2025
A. Mabona, J. Daniel, …, **J. Van Goey**, K. Kalogeropoulos
10.48550/arXiv.2509.24952

A living proteomics benchmark for comprehensive evaluation of deep learning-based *de novo* peptide sequencing tools — Registered report (preprint) 2026
M. Pominova, **J. Van Goey**, C. Adams, …, W. Bittremieux
10.6084/m9.figshare.31680883.v1

Generalizable direct protein sequencing with InstaNexus — Mol. Cell. Proteomics 2026
M. Reverenna, M. Wennekers Nielsen, …, **J. Van Goey**, …, K. Kalogeropoulos
10.1016/j.mcpro.2026.101547

A Framework for Database Search with AI Models in Mass Spectrometry-Based Proteomics — Journal of Proteome Research 2026
K. Kalogeropoulos, **J. Van Goey**, T. P. Jenkins, K. M. Eloff
10.1021/acs.jproteome.5c01153

Education

HOWEST Hogeschool West-Vlaanderen, Microdegree Machine Learning & Deep Learning — Kortrijk, Belgium, 2018 – 2019

Katholieke Universiteit Leuven, Bioinformatics (partial credits obtained) — Leuven, Belgium, 2010 – 2012

University of Antwerp, M.Sc. Biology — Antwerp, Belgium, 1997 – 2004

Universitat de Barcelona, Erasmus interuniversity exchange — Barcelona, Spain, 2002

Møglestu videregående skole, AFS intercultural exchange program — Lillesand, Norway, 1996 – 1997

Skills

Machine learning: Transformers, diffusion models, large language models (LLMs), supervised & self-supervised learning
Frameworks: PyTorch, TensorFlow / TFX, NumPy, pandas
MLOps: Kubernetes-based ML platforms (AIchor), Docker, CI/CD, workflow orchestration (Airflow, Dagster), experiment tracking (MLflow, Neptune, TensorBoard)
Cloud: Buckets, VMs, Cloud Run (AWS, GCP)
Scale: Distributed GPU training, HPC clusters (InstaDeep's Kyber and EuroHPC's MareNostrum 5)
Bioinformatics: BioPython, BLAST, ClustalW, Snakemake, BioNumerics
Programming languages: Python (primary), R
Spoken languages: Dutch (native), English, French & Norwegian (full professional), German & Spanish (notions)

Projects

Awesome De Novo Peptide Sequencing
A comprehensive, interactive map of the *de novo* peptide sequencing field: algorithms, post-processors, downstream applications, and adjacent tools, deep-learning and classical alike.
- Open source: github.com/BioGeek/awesome_de_novo_peptide_sequencing

Self-driving RC car
Turning a 1/16-scale RC car into an autonomous vehicle as a hands-on testbed for modern edge ML: training policies in simulation and deploying them to a Jetson Nano. An ongoing build, documented in a public progress log.
- Progress log: jeroen.vangoey.be/writing (jetson)

SchaakStudieSpinselS 2
Digitised the second volume of my grandfather Ignace Vandecasteele's third book of chess endgame studies, more than three hundred studies, put the whole collection online, and made the chess boards interactive.
- Open source: github.com/BioGeek/SchaakStudieSpinsels2

Photography portfolio
A portfolio of portrait, wedding, maternity, boudoir, landscape and long-exposure photography.

Teaching & mentoring

AIMS South Africa MSc supervision — AIMS South Africa, with InstaDeep — 2025 – 2026

Co-supervised several AIMS South Africa Master's students at InstaDeep on de novo peptide sequencing and proteomics, including InstaNovo interpretability (sparse autoencoders), diffusion models for mass spectrometry, glyco-peptide fine-tuning of InstaNovo, and AI-native proteomic database search via contrastive joint embeddings and a transformer-based scoring function.

DTU PhD research stay — InstaDeep, Cape Town, South Africa — 2026

Mentored a Technical University of Denmark (DTU) PhD student during a 3-month research stay at InstaDeep Cape Town, developing a foundational model for proteomics with rigorous software-engineering practices, part of the ongoing DTU and InstaDeep collaboration.

Machine Learning for Biology practical — Deep Learning Indaba, Ghana — 2023

Mentored the hands-on practical alongside InstaDeep and Google DeepMind colleagues.

Scientific Python training & mentoring — BASF and Bayer Crop Science, Belgium — 2018 – 2020

Trained and mentored researchers and data scientists in scientific Python (NumPy, pandas, BioPython) as manager of the Python/R analysis platform.

Hackathons (co-created, organized and judged)

2025: Machine Learning Interatomic Potentials Hackathon, IndabaX (Stellenbosch University, South Africa)
2024: Snakes and Sequences: Senegalese Serpent Venom Sequencing Hackathon, Deep Learning Indaba (Senegal)
2024: Desert Locust Breeding Ground Prediction Hackathon, IndabaX (Wits, Johannesburg)
2023: Unveiling Cassava's Secrets, Deep Learning Indaba (Ghana)